

1. Project Vision and Stakeholders

1.1 Project Vision

To design and develop a web-based Satellite Data Visualization Platform that transforms complex, raw satellite imagery from publicly available sources such as NASA and NOAA into interactive, easy-to-understand visual insights. The platform will empower meteorologists, researchers, disaster management agencies, agricultural planners, students, and policymakers to identify weather patterns, track climate trends, and detect anomalies such as cyclones, droughts, and temperature spikes without requiring specialized scientific software or deep technical expertise. By democratizing access to satellite-derived climate data, the platform aims to accelerate decision-making across disaster preparedness, agriculture, urban planning, and environmental monitoring, while providing a scalable foundation for future predictive analytics.

1.2 Project Objectives

- Build a reliable data ingestion pipeline that retrieves and processes satellite imagery from NASA and NOAA sources.
- Provide an interactive, map-based interface for visualizing real-time and historical weather patterns.
- Enable time-series analysis of climate variables such as temperature, rainfall, and cloud cover.
- Automatically detect and highlight anomalies such as cyclone formation, droughts, and temperature spikes.
- Allow overlay and comparison of multiple data layers, such as precipitation against vegetation indices.
- Support export of processed datasets and visualizations for offline research use.
- Offer region-specific filtering so users can focus on localized data of interest.
- Ensure the entire platform is accessible through a simple, low-barrier web interface.

1.3 Key Stakeholders

Stakeholder	Role / Interest
Meteorologists	Primary users who monitor atmospheric conditions and generate weather forecasts using the platform's visualizations.

Stakeholder	Role / Interest
Climate Researchers	Use historical and real-time data for long-term climate trend studies and academic research.
Disaster Management Agencies	Depend on timely anomaly detection (cyclones, floods, droughts) for emergency response planning.
Government / Policy Makers	Use aggregated insights to inform public policy on climate resilience and resource allocation.
Agricultural Planners	Use precipitation and vegetation-index overlays to plan crop cycles and irrigation.
Students / Academic Institutions	Use the low-barrier interface for learning and coursework in climate science.
Product Owner	Represents stakeholder interests, defines and prioritizes the product backlog.
Scrum Master	Facilitates Agile ceremonies and removes impediments for the development team.
Development Team	Designs, builds, tests, and maintains the ingestion pipeline, visualization modules, and UI.
Data Providers (NASA / NOAA)	External sources supplying raw satellite imagery and atmospheric datasets.

2. Epics and User Stories

The product requirements have been organized into six Epics, each broken down into detailed User Stories following the standard Agile format: "As a <user>, I want <feature>, so that <benefit>."

EPIC-1: Satellite Data Ingestion and Processing

Retrieve, process, and store raw satellite imagery from public sources (NASA, NOAA) so that clean, structured data is available for visualization.

US-01: As a researcher, I want the system to automatically fetch the latest satellite imagery from NASA and NOAA feeds, so that I don't have to manually download raw data files every day.

US-02: As a system administrator, I want raw satellite data to be automatically converted into a standardized processed format, so that downstream visualization modules can consume it without extra manual processing.

US-03: As a meteorologist, I want the ingestion pipeline to run on a scheduled basis, so that I always have access to near real-time satellite updates.

US-04: As a developer, I want ingestion errors and missing data feeds to be logged and flagged, so that data quality issues are caught before they affect analysis.

EPIC-2: Interactive Map-Based Visualization

Provide an interactive map interface displaying real-time and historical weather patterns.

US-05: As a meteorologist, I want to view real-time satellite imagery overlaid on an interactive world map, so that I can quickly assess current atmospheric conditions.

US-06: As a researcher, I want to zoom and pan across the map to focus on a specific geographic area, so that I can study weather patterns at a regional or local scale.

US-07: As a disaster management officer, I want to toggle between infrared, visible, and radar imagery layers on the map, so that I can compare different views of the same weather event.

US-08: As a user, I want to select a historical date and view the corresponding satellite map, so that I can study how a weather system evolved over time.

EPIC-3: Time-Series Climate Trend Analysis

Track climate variables such as temperature, rainfall, and cloud cover over selected time periods using charts.

US-09: As a climate researcher, I want to view a time-series chart of temperature for a selected region, so that I can analyze long-term warming or cooling trends.

US-10: As an agricultural planner, I want to view rainfall trends over a custom date range, so that I can plan crop cycles and irrigation schedules accordingly.

US-11: As a user, I want to compare cloud cover trends across multiple years, so that I can identify seasonal or long-term patterns.

US-12: As a researcher, I want to export a time-series chart as an image or CSV, so that I can include it directly in my research reports.

EPIC-4: Anomaly and Pattern Detection

Automatically highlight unusual weather patterns such as cyclone formation, droughts, or temperature spikes.

US-13: As a disaster management agency, I want the system to automatically flag potential cyclone formation on the map, so that I can initiate early warning and preparedness measures.

US-14: As a agricultural planner, I want the system to highlight drought-prone regions based on rainfall deficits, so that I can advise farmers on water conservation early.

US-15: As a researcher, I want to receive a notification when a temperature anomaly is detected in a monitored region, so that I can investigate significant deviations promptly.

US-16: As a user, I want anomalies to be visually distinguished with clear map markers, so that I can immediately identify areas needing attention.

EPIC-5: Comparative Multi-Layer Visualization and Export

Allow overlay of multiple data layers and export of processed data and visualizations.

US-17: As a researcher, I want to overlay precipitation data with vegetation index data on the same map, so that I can study the relationship between rainfall and crop health.

US-18: As a user, I want to adjust the transparency of each overlaid data layer, so that I can clearly compare overlapping information.

US-19: As a researcher, I want to download processed satellite datasets in common formats (CSV, GeoTIFF), so that I can perform further analysis in external tools.

US-20: As a planner, I want to export a generated map view as an image file, so that I can include it in reports and presentations.

EPIC-6: Region Filtering and Accessible Web Interface

Provide localized filtering and ensure the whole platform is usable by non-technical stakeholders through a simple web interface.

US-21: As a local government official, I want to filter data by state, district, or custom-drawn region, so that I can focus only on the area under my jurisdiction.

US-22: As a student, I want a simple, guided web interface with tooltips, so that I can use the platform without prior technical training.

US-23: As a user, I want the platform to work smoothly on both desktop and tablet browsers, so that I can access data visualizations from the field or office.

US-24: As a policymaker, I want a dashboard summarizing key regional climate indicators at a glance, so that I can make quick, informed decisions without deep technical analysis.

3. Acceptance Criteria

Acceptance criteria for the highest-priority user stories selected for Sprint 1 are defined below using the Given–When–Then format.

US-01: Automatic satellite data fetching

Given: a scheduled ingestion job is configured for NASA/NOAA feeds

When: the scheduled time is reached and the source feed is available

Then: the system retrieves the latest imagery files and stores them in the raw data repository without manual intervention

US-05: Real-time imagery on interactive map

Given: a user opens the visualization dashboard

When: real-time satellite data is available for the selected region

Then: the map displays the latest imagery overlay within 5 seconds, with a visible timestamp of the last update

US-08: Historical map view by date

Given: a user is on the map view and selects a historical date from the date picker

When: the user confirms the selection

Then: the system loads and displays the satellite imagery corresponding to that exact date and region

US-09: Temperature time-series chart

Given: a user selects a region and a date range

When: the user requests the temperature trend view

Then: a line chart is displayed showing temperature values across the selected range, with axis labels and units clearly indicated

US-13: Automated cyclone flagging

Given: incoming satellite data shows atmospheric patterns matching cyclone-formation criteria

When: the anomaly detection engine processes the latest data batch

Then: the system marks the affected region on the map with a distinct cyclone-warning icon and logs the event with a timestamp

US-17: Multi-layer data overlay

Given: a user has selected precipitation data as the base layer

When: the user enables the vegetation index layer

Then: both layers are rendered simultaneously on the same map with independent opacity controls

US-19: Dataset export

Given: a user has generated a processed dataset view

When: the user clicks the export button and selects a format (CSV or GeoTIFF)

Then: the system generates and downloads a correctly formatted file containing the selected data within 10 seconds

US-21: Region-specific filtering

Given: a user is on the dashboard

When: the user selects a state or district from the region filter or draws a custom boundary

Then: all visualizations on the dashboard update to display data only for the selected region

4. Product Backlog Prioritization

The complete product backlog below is prioritized using the MoSCoW technique (Must have, Should have, Could have, Won't have this release), combined with a business-value rating and dependency mapping.

Story ID	User Story (short)	MoSCoW Priority	Business Value	Dependency
US-01	Automatic satellite data fetching	Must Have	High	None
US-05	Real-time imagery on map	Must Have	High	US-01
US-02	Standardized data processing	Must Have	High	US-01
US-13	Automated cyclone flagging	Must Have	High	US-01, US-02
US-09	Temperature time-series chart	Must Have	High	US-02
US-21	Region-specific filtering	Must Have	Medium	US-05
US-08	Historical map view by date	Should Have	Medium	US-05
US-17	Multi-layer data overlay	Should Have	Medium	US-05
US-19	Dataset export (CSV/GeoTIFF)	Should Have	Medium	US-02
US-14	Drought region highlighting	Should Have	Medium	US-02
US-10	Rainfall trend chart	Should Have	Medium	US-02
US-06	Map zoom and pan	Should Have	Medium	US-05
US-07	Toggle imagery layer types	Should Have	Medium	US-05
US-15	Temperature anomaly notification	Could Have	Low	US-13
US-16	Visual anomaly markers	Could Have	Low	US-13
US-11	Multi-year cloud cover comparison	Could Have	Low	US-09
US-12	Export chart as image/CSV	Could Have	Low	US-09
US-18	Layer transparency control	Could Have	Low	US-17
US-20	Export map view as image	Could Have	Low	US-05
US-03	Scheduled ingestion pipeline	Must Have	High	US-01
US-04	Ingestion error logging	Should Have	Medium	US-01

Story ID	User Story (short)	MoSCoW Priority	Business Value	Dependency
US-22	Guided web interface with tooltips	Should Have	Medium	None
US-23	Cross-device responsive support	Could Have	Low	None
US-24	Regional summary dashboard	Won't Have (this release)	Low	US-21

5. Sprint Backlog

The following high-priority user stories from the product backlog were selected for Sprint 1, based on business value and dependency order. Each story is broken down into actionable development tasks.

US-01: Automatic satellite data fetching

- Identify and document NASA/NOAA public API endpoints
- Build a connector service to authenticate and fetch imagery files
- Implement scheduled job (cron-based) to trigger fetch operations
- Store raw files in a versioned data lake / repository

US-02: Standardized data processing

- Define a common internal data schema for imagery metadata
- Build a transformation service converting raw formats to the standard schema
- Write unit tests validating transformed output structure

US-05: Real-time imagery on interactive map

- Set up map rendering library (e.g., Leaflet/Mapbox) in frontend
- Build API endpoint serving latest processed imagery tiles
- Implement map overlay component with auto-refresh
- Add last-updated timestamp indicator on the UI

US-09: Temperature time-series chart

- Build API endpoint aggregating temperature data by region and date range
- Implement charting component (line chart) in frontend
- Add region and date-range selector controls

US-13: Automated cyclone flagging

- Define rule-based anomaly-detection criteria for cyclone signatures
- Implement detection service running on each ingestion batch
- Render warning markers on the map for flagged regions
- Log detected anomaly events with timestamp and region metadata

6. Story Point Estimation

Story points were estimated using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13...), reflecting relative effort, complexity, and uncertainty rather than absolute time.

Story ID	User Story (short)	Story Points (Fibonacci)	Justification
US-01	Automatic satellite data fetching	5	Requires external API integration, authentication, and scheduling logic; moderate complexity and some uncertainty around third-party API reliability.
US-02	Standardized data processing	8	Involves designing a transformation pipeline across multiple imagery formats; higher complexity and testing effort.
US-05	Real-time imagery on interactive map	5	Requires frontend map integration and backend tile-serving API; well-understood but moderate integration effort.
US-09	Temperature time-series chart	3	Standard charting feature with a defined data aggregation endpoint; low-to-moderate complexity.
US-13	Automated cyclone flagging	8	Requires anomaly-detection rules, backend processing, and map marker rendering; higher uncertainty and logic complexity.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

By the end of Sprint 1, deliver a working prototype of the Satellite Data Visualization Platform that automatically ingests and standardizes satellite imagery from NASA/NOAA, displays it on a real-time interactive map, provides basic temperature time-series analysis, flags cyclone-formation anomalies, and allows users to filter all visualizations by region — establishing the core data pipeline and visualization foundation for future sprints.

7.2 Expected Sprint Deliverables

- A functioning data ingestion connector that automatically fetches satellite imagery from NASA/NOAA on a scheduled basis.
- A data transformation service that converts raw imagery into a standardized, queryable format.
- An interactive web-based map displaying real-time satellite imagery with an auto-refresh mechanism.
- A temperature time-series chart with region and date-range selection controls.
- An automated cyclone-detection feature that visually flags at-risk regions on the map.
- A region-specific filtering control (state/district/custom area) applied across all dashboard visualizations.
- A total estimated velocity of 34 story points delivered and demonstrated in the Sprint Review.

7.3 Definition of Done

- Code is peer-reviewed, merged, and passes all automated unit and integration tests.
- Each delivered user story satisfies its documented acceptance criteria.
- The feature is deployed to a staging environment and verified by the Product Owner.
- Relevant documentation (API endpoints, data schema) is updated.